

 **Reading:** Course Resources
10 min

 **Reading:** How to Use Discussion Forums
10 min

 **Discussion Prompt:**
Introduce yourself
10 min

 **Reading:** Earn a Course Certificate
10 min

 Reading: Are you interested in earning an online MSEE degree?
10 min

3.1.1: Welcome to the course!

3.1.2: Introducing nonlinear variations to Kalman filters

3.1.3: Deriving the three extended-Kalman-filter prediction steps

3.1.4: Deriving the three extended-Kalman-filter correction steps

3.1.5: Introducing a nontrivial EKF example, finding derivatives

3.1.6: Introducing Octave code to initialize and control EKF for state estimation

3.1.7: Introducing Octave code to update EKF for state estimation

3.1.8: Summary of "The extended Kalman filter" module plus next steps

You may be interested to know that the University of Colorado at Boulder (CU-Boulder) offers a completely online Master of Science in Electrical Engineering (MSEE) degree program (<https://www.colorado.edu/ecee/msee>). Any work that you have already completed in the Coursera non-degree certificate “Applied Kalman Filtering” can be transferred into this accredited graduate-degree-earning program.

You may be interested to know that the University of Colorado at Boulder (CU-Boulder) offers a completely online Master of Science in Electrical Engineering (MSEE) degree program (<https://www.colorado.edu/ecee/msee>). Any work that you have already completed in the Coursera non-degree certificate “Applied Kalman Filtering” can be transferred into this accredited graduate-degree-earning program.

The primary additional requirements of the CU-Boulder program beyond the non-degree Coursera course(s) in which you have already enrolled are:

- To receive credit at CU-Boulder, you must pay tuition to CU-Boulder. The tuition fees for the online MSE are substantially lower than for an in-person MSE. There are no financial-aid possibilities.
- You must complete all graded assessments for each Coursera course
- You must complete an online two-hour final examination for each Coursera course. (These are similar to a long weekly quiz and are designed to assess your mastery of the content for the entire course; each of my courses also has a practice exam for you to attempt before taking the final exam.)

There are further requirements on acceptable grades and so forth, and these are listed on the CU-Boulder website. Note that learners enrolled in the CU-Boulder graduate-degree-earning program benefit from access to CU-Boulder teaching assistants who hold regular "office hours" to help answering questions related to course content.

For each of the courses in the Applied Kalman Filtering specialization, you can receive a varying amount of graduate credit. The following list gives the CU-Boulder course number and amount of credit for each course. The MSEE degree requires a total of 30 credits.

- ECEA 5850 Kalman-filter boot camp and state-estimation application (0.8 credits)
- ECEA 5851 Kalman filter deep dive and target-tracking application (0.8 credits)
- ECEA 5852 Nonlinear Kalman filters, parameter-estimation application (0.8 credits)
- ECEA 5853 Particle filters and navigation application (0.8 credits)

Beyond these courses, CU-Boulder offers specializations in embedded systems; power electronics; photonics and optics; and has a roadmap of planned specializations in control and communications; electromagnetics, RF, microwaves, and remote sensing; and computer engineering. The course list is strong and growing.

If this MSEE opportunity interests you, I recommend that you take some time to explore the CU-Boulder website for the program. You can request more information through <https://www.coursera.org/degrees/msee-boulder#request-info>. If you have questions, please reach out to msee-moc@colorado.edu. Note that I am only the content provider for my courses and will not be able to help you with admissions questions.

[Go to next item](#)

✓ Completed